Software Test Plan

Hostel Management System (HMS)

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Revision history

Version	Date	Author	Change summary	Status
1.0	16-09-2026	Team 1	Initial Software Test Plan, written with the SRS and SAD	Draft
1.1	08-11-2026	Team 1	Security validation moved to section 5.1; planned and actual environment and schedule; test cases updated to what was built (31 cases, with status); results section and links to the Unit Test, System Validation, Code Coverage and Branch Coverage reports; approvals	Final

1. Introduction

Purpose: This document defines the test plan for HMS v1.0. It establishes the testing objectives, scope, approach, test environment, resources, schedule, deliverables, responsibilities, risks, traceability and acceptance criteria. Version 1.1 also records what was tested and the results.

Scope: Testing covers the HMS residency lifecycle: admission and allocation, fees and payments, gate/outpass and attendance, mess, complaints, visitors, administration/reporting, security, performance, availability, backup/recovery, usability and accessibility. The SRS defines these functional and non-functional requirements and their acceptance tests.

References: Team 1 Software Requirements Specification (SRS), Team 1 Software Architecture Document (SAD), instructor Software Test Plan template, OWASP-based security validation, PCI-DSS payment-handling constraint, WCAG 2.1 AA accessibility target, and the project's `DEVIATIONS.md` and `docs/reports/`.

Definitions: HMS — Hostel Management System; SRS — Software Requirements Specification; SAD — Software Architecture Document; RTM — Requirements Traceability Matrix; RBAC — Role-Based Access Control; MFA — Multi-Factor Authentication; UAT — User Acceptance Testing; SLA — Service Level Agreement; RPO/RTO — Recovery Point/Recovery Time Objectives.

2. Test Items

- Resident Portal — admission application, allocation status, fees/receipts, outpass, mess opt-out and complaints.
- Staff Back-Office — warden, security guard, maintenance staff, mess manager and administrator workflows.
- Gate Terminal Client — RFID/biometric entry-exit capture through the keyed device API, manual fallback on the guard's Gate log page. Local buffering and ordered replay were not built in the browser (Deviation 15); the API keeps each event's original time and ignores repeats, so a client that buffers can replay safely.
- Auth & RBAC Service — authentication, authorization, session expiry and privileged MFA.
- Admission & Allocation Service — eligibility, bed allocation and transfer.
- Fee & Payment Service — invoice, payment, receipt and overdue processing.
- Gate & Outpass Service — movement records, outpass, curfew alerts and night attendance.
- Mess Service — menu, opt-out and consumption-based billing.
- Complaint Service — lodging, assignment, lifecycle and SLA escalation.
- Visitor Service — visitor registration and visiting-hours enforcement.
- Reporting & Admin Service — reports, room inventory, no-dues and vacating/refund.
- Audit Service and PostgreSQL data layer — auditability, integrity and data retention.

3. Features to be Tested

Area	**SRS IDs**	**Primary acceptance/test cases**
Admission & Allocation	HMS-F-001 to F-003; F-061	TC-ADM-01, TC-ALL-01, TC-ALL-02, TC-INV-01
Fees & Payments	HMS-F-010 to F-013	TC-FEE-01 to TC-FEE-04
Gate / Outpass / Attendance	HMS-F-020 to F-023	TC-GATE-01, TC-GATE-02, TC-OUT-01, TC-ATT-01
Mess	HMS-F-030, F-031	TC-MESS-01, TC-MESS-02
Complaints	HMS-F-040 to F-042	TC-CMP-01 to TC-CMP-03
Visitors / Vacating	HMS-F-050, F-062	TC-VIS-01, TC-VAC-01

Area	SRS IDs	Primary acceptance/test cases
Reporting / Operations	HMS-F-060; NF-001 to NF-005	TC-RPT-01, TC-PERF-01, TC-OPS-AVAIL, TC-OPS-01, TC-OPS-02
Security	HMS-SR-001 to SR-006	TC-SEC-01 to TC-SEC-06
Usability / Accessibility	HMS-NF-004	TC-UX-01

TC-OPS-AVAIL is new in version 1.1: the availability requirement (HMS-NF-002) was listed as "operations evidence" in version 1.0 and now has its own case.

4. Features not to be Tested

- Internal implementation of the institutional ERP; HMS only tests the documented read-only integration interface (a stub is used, Deviation 1).
- Internal implementation of the payment gateway; HMS tests its integration, signed callback handling and the requirement that cardholder data is not stored (a bundled mock gateway is used, Deviation 3).
- Internal implementation of the SMS/email provider; HMS tests notification requests, retry behaviour and failure handling at the interface (email only, Deviation 7).
- Internal firmware/design of the RFID or biometric reader; HMS tests the device event interface and gate-side handling (Deviation 8).
- Third-party systems beyond the interfaces exposed to HMS, consistent with the SAD scope.

5. Test Approach / Strategy

5.1 Security Validation

- Verify TLS 1.2 or higher and reject weak/unsupported configurations.
- Test every role against restricted functions; unauthorized requests must return HTTP 403 where applicable.
- Verify 15-minute idle session expiry and token invalidation after logout.
- Verify Warden and Administrator MFA and salted password hashing using Argon2 (as specified by the SAD).
- Verify append-only/tamper-evident audit records for allocation, fee, outpass approval, override and account-management actions.
- Verify card numbers, CVV and bank credentials are not stored in the HMS database or logs.
- Perform negative/fuzz input tests and OWASP-based checks for SQL injection and XSS.
- Verify sensitive resident, guardian, medical and identity-proof data is not exposed through unauthorized screens or API responses.

Outcome: cases TC-SEC-02 to TC-SEC-05 pass. TC-SEC-01 (live TLS scan) and TC-SEC-06 (OWASP ZAP scan) are partial: the controls are enforced and tested, but no deployed server or ZAP run was available (Deviation 17).

5.2 Test Levels

- Unit testing — individual Django app modules, models, serializers, validation and business rules.
- Integration testing — REST APIs, PostgreSQL data access, ERP, payment, notification and gate-reader interfaces.
- System testing — end-to-end resident and staff workflows across the modular monolith.
- Acceptance testing / UAT — verification against the SRS acceptance criteria and representative hostel workflows.

5.3 Test Types

- Functional testing — verify all High/Medium/Low functional requirements.
- Regression testing — rerun impacted tests after fixes and changes (the full suite runs on every pull request in GitHub Actions).
- API testing — validate request/response contracts, authentication, authorization and error envelopes.
- Performance and load testing — verify the 2-second 95th-percentile target at 200 concurrent users with 2,000 seeded resident records.
- Security testing — TLS, RBAC, session expiry, MFA, audit logging, payment-data handling, input validation, SQL injection/XSS and OWASP-based vulnerability scanning.
- Usability and accessibility testing — mobile layout, keyboard navigation, WCAG 2.1 AA and task completion for low-skill users.
- Reliability/recovery testing — uptime evidence, nightly backup and restore drill.
- Compatibility testing — current Chrome, Edge, Firefox and Safari; desktop and Android/iOS mobile browsers (executed in Chrome only, Deviation 17).

5.4 Entry and Exit Criteria

Entry criteria: Stable build available; required modules deployed; test database and representative data available; test environment configured; external-system mocks/sandboxes or accessible interfaces available; test cases reviewed.

Exit criteria: 100% of planned test cases executed; every High-priority functional requirement implemented and verified; no non-functional or security requirement fails; all High-severity defects are closed; RTM shows Pass for every required test case.

Assessment against the exit criteria (8 November 2026):

Criterion	Met?
100% of planned cases executed	No: 30 of 31 (TC-OPS-AVAIL needs a month of uptime data)
Every High-priority functional requirement implemented and verified	Yes: all 20 functional requirements pass
No non-functional or security requirement fails	Yes: none fails; five are partial or not run
All High-severity defects closed	Yes: 7 High defects, all fixed and retested
RTM shows Pass for every required case	No: 26 of 31 (five need evidence the project cannot produce, see section 17)

6. Test Environment

Category	Planned environment	Actual environment
Backend	Python 3.12, Django + Django REST Framework	Python 3.12 in CI (3.13 on developer machines), Django 5.2, DRF 3.16
Frontend	Django templates, Bootstrap 5, server-rendered responsive pages	As planned
Gate client	Server-rendered page + vanilla JS + local IndexedDB buffer	Server-rendered Gate log page and keyed device API; no browser buffer (Deviation 15)
Database	PostgreSQL through Django ORM/migrations	PostgreSQL 16 locally and in CI

Category	Planned environment	Actual environment
External interfaces	ERP sandbox or stub; PCI-DSS gateway sandbox/stub; SMS/email test service; RFID/biometric event stub	ERP stub, bundled mock gateway with signed callbacks, Django mail backend (console in development), keyed device event API
Browsers/devices	Chrome, Edge, Firefox, Safari; desktop; Android/iOS; gate workstation	Chrome on desktop; every page also checked at 360 px (Deviation 17)
Performance	JMeter or equivalent; 200 concurrent users, 2,000 seeded records	Locust, 200 users, 2,000 seeded residents, on a developer laptop with Waitress (Deviation 14)
API / UI tools	Postman/OpenAPI-style tests; Selenium where useful	Django and DRF test clients in CI; axe-core accessibility audit in Chrome (Deviation 16)
Defect tracking	Jira	GitHub issues and project board; defect table in `docs/qa/TEST_SUMMARY.md` (Deviation 16)
Reporting	PDF/XLSX generation validation	ReportLab PDFs and openpyxl workbooks, opened and checked in the tests

7. Test Schedule

Milestone	Planned	Actual
Test plan preparation and review	16 Sep 2026	16 Sep 2026
Test case design and RTM verification	17–18 Sep 2026	Sep 2026
Test environment and test-data setup	18–19 Sep 2026	Oct 2026 (demo data and CI set up in Sprint 1)
Unit and integration testing	20–23 Sep 2026	19 Oct – 2 Nov 2026, written with each feature in the sprints
System / functional test execution	24–28 Sep 2026	2 Nov 2026 (QA register)
Security, performance and accessibility testing	29 Sep – 01 Oct 2026	27 Oct – 2 Nov 2026
Regression testing and defect verification	02–03 Oct 2026	2 and 8 Nov 2026
UAT / acceptance review	04–05 Oct 2026	2 Nov 2026 (role walkthrough and Sprint 2 demo video)
Test summary and sign-off	06 Oct 2026	8 Nov 2026 (reports in `docs/reports/`)

The planned dates assumed the build would be ready in September. Implementation ran in the October sprints, so test execution moved after the build (Deviation 18).

8. Test Deliverables

Deliverable	Where it is
Software Test Plan (this document)	Repository root
Test cases (functional, API, security, performance, usability), updated	`docs/reports/TEST_CASE_DOCUMENT.md`; section 15
Automated test scripts	`apps/*/test*.py`, `hms/test_*.py`; 376 tests run by `python manage.py test`
Test data and seeded database	`manage.py seed_demo`; Locust data generators
Test execution logs and evidence	`docs/reports/unit-test-run.txt`, `docs/qa/`, `docs/performance/`
Defect reports and retest results	`docs/qa/TEST_SUMMARY.md` (26 defects, all closed)
Requirements Traceability Matrix with status	SRS section 8; section 13 below
Unit Test Report	`docs/reports/UNIT_TEST_REPORT.md`
System Validation Report (final test summary)	`docs/reports/SYSTEM_VALIDATION_REPORT.md`
Code Coverage Report	`docs/reports/CODE_COVERAGE_REPORT.md`, `coverage-html/`
Branch Coverage Report	`docs/reports/BRANCH_COVERAGE_REPORT.md`

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Vaishnavi Goyal	Maintain the test plan, coordinate execution, monitor coverage and prepare the final summary.
Test Engineer	Yellala Sai Aneesh Reddy	Design and execute test cases, API/UI/security tests, capture evidence and log defects.
Developer	Vignesh Nunna	Provide builds, support environment setup, triage defects and implement fixes.
Product Owner	Vishnu S R	Validate acceptance criteria, review UAT results and support sign-off.

10. Risks and Mitigation

Risk	Impact	Mitigation	Outcome
Unstable or delayed build	Test execution blocked	Request early smoke builds; prioritise critical-path tests first	The build finished late (end of Sprint 2); CI ran the suite on every PR throughout

Risk	Impact	Mitigation	Outcome
ERP/payment/notification dependency unavailable	Integration tests delayed	Use mocks/sandboxes and contract tests	Stub ERP, mock gateway and mail backend used (Deviations 1, 3, 7)
Gate reader/device SDK unavailable	Gate tests blocked	Use a test event stub	Keyed device API tested with simulated events (Deviation 8)
Network outage at gate	Entry/exit events may be lost	Test IndexedDB buffering and ordered replay	Browser buffer not built; the API accepts replayed events safely (Deviation 15)
Peak-season load	Response-time target may fail	Load test 200 users, 2,000 records	p95 1.6 s on a laptop; not repeated on a production-like server
Sensitive data exposure	Security/privacy impact	Synthetic data; RBAC/TLS/encryption/logging controls; security tests	Synthetic data only; access-matrix and input tests pass; ID numbers not encrypted at rest (Deviation 12)
Database failure	Data loss/outage	Verify nightly backups and quarterly restore against RPO 24 h / RTO 4 h	One restore drill: backup 0.9 s, restore 0.7 s, data identical

11. Assumptions & Dependencies

- The institutional ERP provides a stable read-only identity lookup interface or suitable test stub.
- The payment gateway provides a sandbox/test callback mechanism.
- SMS/email test delivery or a mock notification service is available.
- A gate-reader API stub/SDK is available when physical hardware is unavailable.
- Synthetic resident, guardian, fee, gate and complaint data will be used for testing.
- PostgreSQL test data can be reset between test runs.
- Required test tools and browsers are available to the team.
- The project will retain the architectural choices documented in the SAD unless formally revised; differences are recorded in `DEVIATIONS.md`.

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment is unavailable for more than 4 hours or cannot support the planned test run.
- The build is sufficiently unstable that more than 30% of planned tests are blocked.
- A critical defect causes data corruption, security compromise or prevents execution of major workflows.
- Required test dependencies are unavailable and no approved stub/workaround exists.

Resume testing if:

- Blocking/critical defects are resolved or an approved workaround is available.
- The environment is stabilised and test data is restored.
- Required external interfaces or test stubs are available.
- A smoke test confirms the build is suitable for continued execution.

No suspension was needed.

13. Test Case Management & Traceability

The RTM in the SRS (section 8) maps each requirement to a module and test case. The test plan uses the same requirement and test identifiers so that implementation, testing and acceptance remain traceable. Status as of 8 November 2026 ("Dev." is the row in `DEVIATIONS.md`):

Requirement	Title	Test case	Result	Dev.
HMS-F-001	Submit admission application	TC-ADM-01	PASS	—
HMS-F-002	Automatic bed allocation	TC-ALL-01	PASS	—
HMS-F-003	Room transfer by warden	TC-ALL-02	PASS	—
HMS-F-010	Generate fee invoice	TC-FEE-01	PASS	—
HMS-F-011	Online fee payment	TC-FEE-02	PASS	3
HMS-F-012	Receipt generation	TC-FEE-03	PASS	—
HMS-F-013	Overdue flag and reminders	TC-FEE-04	PASS	4, 7
HMS-F-020	Gate entry-exit capture	TC-GATE-01	PASS	8, 15
HMS-F-021	Outpass workflow and guardian notice	TC-OUT-01	PASS	7
HMS-F-022	Curfew violation alert	TC-GATE-02	PASS	9
HMS-F-023	Night-attendance register	TC-ATT-01	PASS	—
HMS-F-030	Publish mess menu	TC-MESS-01	PASS	10
HMS-F-031	Mess opt-out and billing	TC-MESS-02	PASS	10
HMS-F-040	Lodge complaint	TC-CMP-01	PASS	—
HMS-F-041	Complaint lifecycle	TC-CMP-02	PASS	—
HMS-F-042	SLA escalation	TC-CMP-03	PASS	11
HMS-F-050	Visitor registration and hours	TC-VIS-01	PASS	12
HMS-F-060	Management reports	TC-RPT-01	PASS	13
HMS-F-061	Room inventory master	TC-INV-01	PASS	—
HMS-F-062	No-dues clearance and refund	TC-VAC-01	PASS	—
HMS-NF-001	Response time and capacity	TC-PERF-01	PARTIAL	14
HMS-NF-002	Availability 99.5%	TC-OPS-AVAIL	NOT RUN	14
HMS-NF-003	Backup and recovery	TC-OPS-01	PASS	14
HMS-NF-004	Usability and accessibility	TC-UX-01	PARTIAL	17
HMS-NF-005	Data retention	TC-OPS-02	PASS	—
HMS-SR-001	TLS 1.2 or higher	TC-SEC-01	PARTIAL	6, 17
HMS-SR-002	RBAC and session expiry	TC-SEC-02	PASS	—
HMS-SR-003	Password hashing and MFA	TC-SEC-03	PASS	—
HMS-SR-004	Audit logging	TC-SEC-04	PASS	—

Requirement	Title	Test case	Result	Dev.
HMS-SR-005	No cardholder data stored	TC-SEC-05	PASS	—
HMS-SR-006	Input validation and injection defence	TC-SEC-06	PARTIAL	17

14. Test Metrics & Reporting

Metrics defined: test case execution percentage; pass/fail/blocked percentage; requirement coverage; defect density and severity distribution; defect aging and retest status; regression pass rate; performance (95th percentile and concurrent users); security findings; UAT completion.

Reports: daily/iteration execution status, defect report, requirement coverage report, performance/security evidence and the final test summary report.

Values at 8 November 2026:

Metric	Value
Test case execution	30 of 31 executed (97%)
Pass / partial / not run	26 / 4 / 1
Requirement coverage	31 of 31 requirements have a case; 30 have executed evidence
Automated tests	376 run, 376 pass; regression pass rate 100% on the final run
Code coverage	95.0% of statements; 82.8% of branches
Defects	26 found (7 High, 10 Medium, 9 Low), 26 fixed, 0 open
Performance	p95 1.6 s at 2,000 records and 200 users, on a laptop
Security findings	0 open; no ZAP report
Accessibility	0 axe-core violations on 33 pages

15. Test Cases

The cases below are the version 1.0 cases brought up to date with what was built. The same list with the automated test for each case, and what changed from version 1.0, is in `docs/reports/TEST_CASE_DOCUMENT.md`.

Test ID	Requirement	Scenario	Expected result	Status
TC-ADM-01	HMS-F-001	Submit a valid admission application, then a second for the same enrolment and session.	Valid submission gets HMS-2026-NNNNN; the duplicate is refused with a message naming the first (page and API 409).	PASS
TC-ALL-01	HMS-F-002	Allocate an approved applicant while vacant beds exist.	Only vacant, in-service beds matching gender, block and room type are offered; ranking (merit, seniority, first come) is followed.	PASS

Test ID	Requirement	Scenario	Expected result	Status
TC-ALL-02	HMS-F-003	Warden transfers a resident to another bed.	Old bed vacant and new bed occupied in one transaction; reason, warden and time are audited.	PASS
TC-FEE-01	HMS-F-010	Allocate a bed; change the fee structure afterwards.	Invoice total equals the sum of its fee heads and does not change when fees change later.	PASS
TC-FEE-02	HMS-F-011	Send success, failed, cancelled, abandoned, tampered and unsigned gateway callbacks.	Success marks the invoice Paid; the rest leave it Unpaid; tampered or unsigned callbacks are refused.	PASS
TC-FEE-03	HMS-F-012	Pay an invoice and download the receipt.	Receipt RCT-YYYY-NNNNNN and a PDF are available in the same request (within 30 s).	PASS
TC-FEE-04	HMS-F-013	Run the overdue job the day after the due date.	Unpaid invoice becomes Overdue; one reminder to the resident with the guardian in BCC.	PASS
TC-GATE-01	HMS-F-020	Post a reader event; record a manual exit and entry on the Gate log; repeat an event.	A timestamped IN/OUT record is created each time; a repeated event is ignored.	PASS
TC-OUT-01	HMS-F-021	Resident requests an outpass; warden approves or rejects; guard records return.	Pending → Approved/Rejected; guardian emailed on approval and on return, not on rejection.	PASS
TC-GATE-02	HMS-F-022	Leave a resident out 31 minutes past curfew and run the alert job.	One alert, the wardens are emailed, and the alert shows on the warden's dashboard.	PASS
TC-ATT-01	HMS-F-023	Generate the night register for a date.	Room-by-room list with Present / On leave / Absent from gate and outpass records.	PASS
TC-MESS-01	HMS-F-030	Mess manager publishes, edits, removes and copies meals; resident opens Meals.	A published meal is on the residents' page immediately.	PASS
TC-MESS-02	HMS-F-031	Skip a meal inside 24 hours and outside it; undo; bill a month.	Late skip refused; undo allowed until 24 h before; bill = Rs 50 × meals not skipped, once per ended month.	PASS
TC-CMP-01	HMS-F-040	Resident reports a Maintenance problem with a photo; try a photo over 5 MB.	Ticket #N with priority High; the large photo is refused; the complaint is in the warden's queue.	PASS
TC-CMP-02	HMS-F-041	Move a complaint Open → In progress → Resolved → Closed.	Closing requires the resident's confirmation; closing an unresolved one is refused.	PASS
TC-CMP-03	HMS-F-042	Leave High and Medium complaints unresolved past the SLA.	Escalation after 48 h / 96 h, every administrator emailed, entry in the audit log.	PASS
TC-VIS-01	HMS-F-050	Register visitors inside and outside visiting hours; warden overrides.	Out-of-hours entry refused; override needs a reason, which is audited; exit time recorded.	PASS

Test ID	Requirement	Scenario	Expected result	Status
TC-RPT-01	HMS-F-060	Open the management reports; export PDF and XLSX.	Figures reconcile with the records; both files open.	PASS
TC-INV-01	HMS-F-061	Administrator deletes a room with an occupied bed; lowers capacity below beds in use.	Both are refused.	PASS
TC-VAC-01	HMS-F-062	Vacate with incomplete and complete clearances.	Bed released only after all three clearances and paid invoices; refund = deposit - dues - damages; a negative result is payable.	PASS
TC-PERF-01	HMS-NF-001	Load test the resident portal with 200 concurrent users and 2,000 records.	95th percentile $\leq$ 2 s.	PARTIAL
TC-OPS-AVAIL	HMS-NF-002	Monitor uptime over a month.	At least 99.5% available.	NOT RUN
TC-OPS-01	HMS-NF-003	Back up the database and restore it into a fresh one.	Restore within 4 h and data loss within 24 h; data identical; audit chain verifies.	PASS
TC-UX-01	HMS-NF-004	Audit every page for WCAG 2.1 AA, at 360 px and by keyboard; trial an untrained warden.	0 violations; 80% of a 10-user trial complete allocation unaided in 5 min.	PARTIAL
TC-OPS-02	HMS-NF-005	Run the retention report and the anonymisation job.	Records older than 5 years are listed; their personal data is blanked and the records stay retrievable.	PASS
TC-SEC-01	HMS-SR-001	Inspect the TLS configuration.	No connection below TLS 1.2 and no weak ciphers.	PARTIAL
TC-SEC-02	HMS-SR-002	Request every restricted URL as every other role; idle for 15 minutes; reuse a logged-out session.	403 for other roles; idle session and logged-out session are rejected.	PASS
TC-SEC-03	HMS-SR-003	Inspect password storage; sign in as warden and administrator.	Non-reversible salted Argon2 hashes; the second step is mandatory.	PASS
TC-SEC-04	HMS-SR-004	Perform privileged actions; try to edit or delete audit rows.	Actor, action, time, target and outcome recorded; entries cannot be changed.	PASS
TC-SEC-05	HMS-SR-005	Post card data to the gateway page; inspect the database and logs.	No card number, CVV or bank credential is stored.	PASS
TC-SEC-06	HMS-SR-006	Run SQL-injection and stored-markup input on every page.	No injection or XSS finding; queries are parameterised.	PARTIAL

16. Acceptance Test Suites

Suite	Coverage	Result
TS-1 Admission and Allocation	TC-ADM-01, TC-ALL-01, TC-ALL-02, TC-INV-01	4 pass
TS-2 Fees and Payments	TC-FEE-01, TC-FEE-02, TC-FEE-03, TC-FEE-04	4 pass
TS-3 Gate, Outpass and Attendance	TC-GATE-01, TC-GATE-02, TC-OUT-01, TC-ATT-01	4 pass
TS-4 Mess	TC-MESS-01, TC-MESS-02	2 pass
TS-5 Complaints	TC-CMP-01, TC-CMP-02, TC-CMP-03	3 pass
TS-6 Visitors and Vacating	TC-VIS-01, TC-VAC-01	2 pass
TS-7 Reporting and Operations	TC-RPT-01, TC-PERF-01, TC-OPS-01, TC-OPS-02, TC-OPS-AVAIL	3 pass, 1 partial, 1 not run
TS-8 Security	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04, TC-SEC-05, TC-SEC-06	4 pass, 2 partial
TS-9 Usability and Accessibility	TC-UX-01	0 pass, 1 partial

17. Test Execution Summary

Execution is documented in `docs/reports/`:

Report	File	Headline
Unit Test Report	`UNIT_TEST_REPORT.md`	376 tests, all pass
System Validation Report	`SYSTEM_VALIDATION_REPORT.md`	26 pass, 4 partial, 1 not run; all 20 functional requirements pass
Code Coverage Report	`CODE_COVERAGE_REPORT.md`	95.0% of statements
Branch Coverage Report	`BRANCH_COVERAGE_REPORT.md`	82.8% of branches

Open items. These cases need evidence that a course project cannot produce, and are recorded in `DEVIATIONS.md`:

Case	Needs
TC-PERF-01	Load test on a production-like server (the laptop result is 1.6 s at the 95th percentile against a 2 s target)
TC-OPS-AVAIL	A month of operation to measure 99.5% availability
TC-UX-01	A 10-user allocation trial with untrained wardens
TC-SEC-01	A live TLS scan of a deployed server
TC-SEC-06	An OWASP ZAP scan

18. Approvals

Role	Name	Signature / Date
QA Lead	Vaishnavi Goyal	
Development Lead	Vignesh Nunna	
Product Owner	Vishnu S R	

Document status: Final, version 1.1, 8 November 2026. It reflects the completed implementation, the executed test cases and the results in `docs/reports/`.